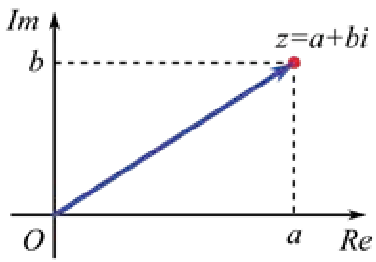
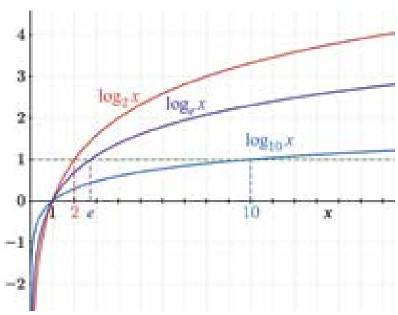
first developed by Italian Mathematician Gerolamo Cardano, were called 'fictitious' at the time. Their name is true to an extent because they represent numbers that do not exist. However, complex numbers play a huge role in physics, chemistry, biology,




economics, electrical engineering and statistics. Complex numbers have two parts – one is real, the other is imaginary. The latter is represented using the letter 'i'.

## LOGARITHMS

Logarithms, in essence, are an extension of exponents. The basic concept behind logarithms is that we can multiply numbers by adding related numbers. They were essential in defining our understanding of radioactive decay and, in the yes-




teryears, made calculations easy for researchers working in astronomy and engineering. They may have, to a large extent, lost their utility at the hands of computers that are much faster in doing calculations. However, they remain an essential part of analogue calculations.

## EULER'S FORMULA FOR POLYHEDRONS

In its crudest sense, Euler's formula for polyhedrons describes the shape

$$F - E + V = 2$$

or structure of a space regardless of its alignment in any direction. Leonhard Euler is said to have been the person who took the first workings of a formula by Descartes, refined proved it before publishing his findings in 1750. This equation has been important in expanding our knowledge of topography and surface geometry. In the modern day, topology helps us decode the behaviour and function of DNA.

## WAVE EQUATION

Based on differential calculus, the wave equation is used to define the behaviour of waves. When it was first developed, it was used to describe the behaviour of a vibrating violin string. As a result of the work of Daniel Ber-

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

noulli and Jean D'Alembert in the 18th century, this equation has helped us better understand sound waves and seismic activities like earthquakes. This equation also helps researchers study ocean activity.

## NAVIER-STOKES EQUATIONS

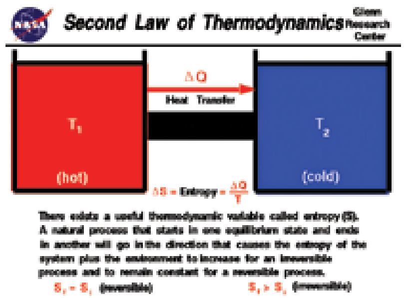
The Navier-Stokes equation is an equation that describes the flow and nature of incompressible fluids. Theoretically, these equations can be solved using the methods of calculus, but in practice, solving these equations analogue has proven to be an impossibility. At one point, these equations were too difficult for even computers to solve, and thus remained untouched. However, once computers became powerful enough to solve them, they have been vastly

$$\rho\left(\frac{\partial v}{\partial t} + v \cdot \nabla v\right) = -\nabla p + \nabla \cdot T + f$$

used, especially in automobile and aircraft aerodynamics.

## THE SECOND LAW OF THERMODYNAMICS

The Second Law of Thermodynamics states that heat and energy dissipate over time. This law, coupled with the concept of entropy, played a pivotal role in developing our understanding of the basic concept of energy. This




Source: NASA

equation has been instrumental in helping understand the concept that in a closed system, when energy changes form or matter moves around, the entropy or disorder of the system increases. This equation led to the development of steam engines.

## THE SCHRÖDINGER EQUATION

This equation helps us predict the future behaviour of a dynamic system and lays the foundation of our understanding of matter as a wave rather than particles. The solution of the Schrödinger equation gives us quantum numbers and an under-

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle$$

standing of the orientations and shapes of orbitals where electrons are found inside a molecule or an atom. It has two variations. One is time-dependent, while the other is time-independent. Erwin Schrödinger was awarded the Nobel Prize in 1933 for developing this equation.

If there are any equations that you feel should have been included, please let us know by dropping a mail to editor@digit.in